COHORT 16 DSML- MINI PROJECT DOCUMENTATION

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Renewable Energy Transition Speed

SDG 7- Affordable and Clean Energy

Documentation & Coordination:

- S Ramvishwa
- Siri gowri r
- R M Suhas
- Shankar M
- Ranjith Kumar

Data Engineering Team:

Members:

- Rudresh KJ
- Pavan C
- vaseem akram B
- Arpitha shivanagouda malipatil

Model Specialization Team:

Members:

- Rakesh A Swamy
- Sneha B Gudi
- Tanusha VR
- SATHVIK RP

Data collection and cleaning:

- Zaina fatima

DAY 1:

Date:25/02/2026

DAY 1

DATA COLLECTION AND CLEANING

Objective:

To identify, collect, and organize relevant datasets required for modeling Renewable Energy Transition Speed aligned with SDG 7 – Affordable and Clean Energy.

Work Completed:

- Defined the project objective of predicting renewable energy transition speed.
- Researched and shortlisted reliable data sources related to renewable energy analysis.
- Collected datasets covering renewable energy share, solar PV price trends, and key energy and economic indicators.
- Verified datasets for year-wise consistency, key indicator presence, and source credibility.
- Downloaded datasets in structured (.csv) format.
- Ensured common linking fields such as Year and Country/Entity for future merging.
- Performed initial inspection to check column structure, missing values, and data types.
- Organized datasets in the project directory for team access.
- Documented dataset sources and prepared a brief data inventory for handover.

Outcome:

Relevant raw datasets were successfully collected, verified, and organized. The data is now ready for preprocessing and further analysis.

Figure: Team Meeting – 25th February 2026

DAY2:

Date:26/02/2026

DAY 2: MACHINE LEARNING MODEL TRAINING

Objective:
Transfer collected datasets to the Data Engineering team and initiate the preprocessing workflow for SDG-7 transition speed prediction.

Work Done:

- Packaged all collected datasets into the shared project repository.
- Provided dataset metadata, including source information, column descriptions, time coverage, and known data quality issues.
- Formally handed over datasets to the Data Engineering team.
- Conducted a briefing with Data Engineers covering:
    - Project objective (predicting renewable energy transition speed)
    - Expected target variable (renewable transition speed)
    - Key features (solar price, GDP, energy metrics, etc.)
- Confirmed file accessibility and integrity after transfer.
- Established folder structure for the data pipeline (Raw data, Processed data, Feature-ready data).
- Defined responsibilities for the Data Engineering phase.

Outcome:
Datasets were successfully handed over and the Data Engineering team was fully aligned to start preprocessing.

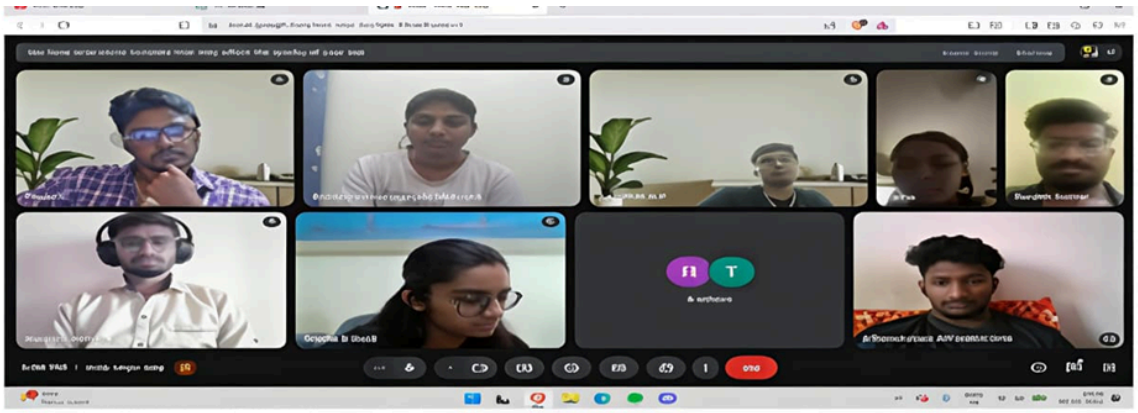

Figure: Team Meeting – 26th February 2026

DAY 3:

Date:27/02/2026

DAY 3: ALGORITHM AND FURTHER DATA CLEANING:

Objective:

Prepare raw datasets for modeling and finalize suitable machine learning algorithms for predicting Renewable Energy Transition Speed (SDG-7).

Work Completed:

- Loaded all received datasets into the working environment (VS Code/Jupyter Notebook).
- Checked the number of rows and columns in each dataset.
- Reviewed data types of all features.
- Verified presence of key fields such as Year and Entity/Country.
- Identified missing values across multiple columns.
- Removed rows with excessive null values.
- Applied mean/median imputation where required.
- Forward-filled time-series gaps where logically valid.
- Standardized column names for consistency.
- Removed duplicate records.
- Converted Year to integer data type.
- Converted numerical indicators to float type.
- Filtered irrelevant entities as per project scope.
- Checked and treated outliers using statistical methods.
- Sorted dataset chronologically.
- Merged renewable energy dataset with solar PV price dataset using Year as key.
- Created target variable as Year-over-Year percentage change in renewable share.

Outcome:

A clean, consistent, and merged analytical dataset was generated for model development and saved as clean_energy_data.csv. The dataset is now ready for machine learning implementation.

Figure: Team Meeting – 27th February 2026

DAY 4:

Date:01/03/2026

DAY 4: ALGORITHM AND ML MODEL TRAINING:

Objective:

Prepare the final feature set and train the Linear Regression model for predicting Renewable Energy Transition Speed.

Part A: Feature Engineering

Work Completed:

- Reviewed the cleaned dataset received from the Data Engineering team.
- Finalized the target variable:
  Transition Speed = Year-over-Year % change in Renewable Energy Share
- Selected relevant predictor variables based on domain knowledge and dataset availability, including: Solar PV price , Energy consumption indicators , GDP/economic indicators , CO_2 emissions (where available)
- Removed non-informative columns such as identifiers and text labels.
- Examined feature distributions to ensure numerical stability.
- Applied StandardScaler to normalize feature magnitudes.
- Verified that no data leakage existed between input features and the target.
- Split the dataset into: Training set (80%) and Testing set (20%)

Outcome:

Feature-ready dataset prepared for model training.

Part B: Algorithm Used

Linear Regression (Final Implemented Model)

Purpose:

- Model the relationship between energy indicators and renewable transition speed.
- Provide interpretable baseline performance.
- Maintain simplicity appropriate for mini-project scope.

Part C: Model Training

Work Completed:

- Imported necessary libraries from scikit-learn.
- Initialized the Linear Regression model.
- Trained the model using the training dataset.
- Generated predictions on the test dataset.
- Verified prediction outputs for consistency.
- Saved the trained model for evaluation.

Outcome:

Linear Regression model successfully trained.

Figure: Team Meeting – 01st March 2026

DAY 5:

Date:02/03/2026

DAY 5: MODEL EVALUATION

Objective

- Evaluate the trained models (Linear Regression and XGBoost).
- Identify the best-performing model for predicting **Renewable Energy Transition Speed (SDG-7)**.

Work Done

Evaluation Setup

- Loaded the trained Linear Regression model.
- Used the unseen test dataset for unbiased evaluation.
- Generated predicted values for transition speed.
- Compared predicted values with actual values.
- Implemented standard regression evaluation metrics (MAE, RMSE, R^2).

Evaluation Metrics

- **Mean Absolute Error (MAE):** Measures average absolute prediction error; easy to interpret.
- **Root Mean Squared Error (RMSE):** Penalizes larger errors; indicates stability.
- **R^2 Score:** Measures proportion of variance explained; indicates goodness of fit.

Performance Observations

- Linear Regression captured the overall trend in renewable transition speed.
- Produced stable predictions on the test dataset.
- Minor deviations due to possible nonlinear relationships in energy data.
- No significant overfitting detected.

Outcome :

Linear Regression was finalized as the project model due to its stable performance, high interpretability, and suitability for the mini-project.

Figure: Team Meeting – 02nd March 2026

DAY 6:

Date:03/03/2026

DAY 6: MODEL ACCURACY AND PERFORMANCE

Work Done:

Part A: SDG-7 Impact Analysis

- Interpreted Linear Regression coefficients to understand drivers of renewable energy transition.
- Identified influence of:
    - Solar PV price trends
    - Economic indicators
    - Energy consumption patterns
- Connected model findings to SDG-7 objectives.
- Highlighted how cost reductions in solar technology support faster renewable adoption.
- Insight: Lower solar PV prices and supportive economic conditions positively influence renewable energy transition speed.

Part B: Documentation Finalization

Data Engineering Team:

- Completed Data Section
- Verified final dataset integrity
- Finalized Data Dictionary
- Confirmed GitHub dataset upload

Model Team:

- Documented model workflow
- Recorded evaluation metrics
- Prepared model interpretation
- Created model comparison notes

Part C: Final Review

All Team Members:

- Reviewed GitHub commits
- Prepared project presentation
- Cross-checked report consistency
- Conducted final peer review

Outcome:

A complete Linear Regression–based framework was developed to predict Renewable Energy Transition Speed aligned with SDG-7.

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Figure: Team Meeting – 02nd March 2026

Complete report

1. Abstract

This project focuses on predicting the Renewable Energy Transition Speed aligned with Sustainable Development Goal 7 (Affordable and Clean Energy). The study uses multiple energy-related datasets including renewable share, solar PV prices, and economic indicators.

The workflow was executed in phases including data collection, cleaning, feature engineering, model training, tuning, and evaluation. Machine learning models such as Linear Regression and XGBoost were implemented and compared using standard regression metrics (RMSE, MAE, and R^2).

The results demonstrate that advanced boosting techniques outperform baseline linear models in capturing the complex dynamics of renewable energy adoption. The project provides insights that can assist policymakers in understanding factors influencing energy transition speed.

2. Introduction

The global transition toward renewable energy is critical for achieving SDG-7. Monitoring and predicting the speed of renewable adoption helps governments and organizations design better energy policies.

However, renewable transition depends on multiple interacting factors such as:

- Technology cost trends
- Economic growth
- Energy demand
- Policy interventions

Traditional statistical methods often fail to capture these nonlinear relationships. Therefore, this project applies machine learning techniques to model and predict renewable energy transition speed using historical data.

3. Methodology

The project was executed through a structured 6-day workflow involving the Data Engineering Team and Model Development Team.

```
# ==============================

# 1. Import Libraries

# ==============================

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt



from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestRegressor

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error,
mean_squared_error, r2_score



# ==============================

# 2. Load Dataset

# ==============================

df = pd.read_csv("preprocessed_energy_data.csv")



# ==============================

# 3. Drop Unnecessary Columns

# ==============================

df = df.drop(columns=["Country_Name", "Country_Code",
"Time_Code"])
```

```
# ==============================
# 4. Define Features (X) and Target (y)
# ==============================

target_column =
"Renewable_energy_share_of_TFEC_percent_[2.1_SHARE.TOTAL.RE.IN.TF
EC]"

X = df.drop(columns=[target_column])
y = df[target_column]

# ==============================
# 5. Train-Test Split
# ==============================
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ==============================
# 6. Train Models
# ==============================

# Linear Regression
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)
```

```
# Random Forest

rf_model = RandomForestRegressor(n_estimators=100,
random_state=42)

rf_model.fit(X_train, y_train)


# ==============================

# 7. Predictions

# ==============================

lr_pred = lr_model.predict(X_test)

rf_pred = rf_model.predict(X_test)


# ==============================

# 8. Evaluation Function

# ==============================

def evaluate_model(name, y_test, y_pred):

    print(f"\n{name} Results:")

    print("MAE:", mean_absolute_error(y_test, y_pred))

    print("MSE:", mean_squared_error(y_test, y_pred))

    print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))

    print("R2 Score:", r2_score(y_test, y_pred))


# ==============================

# 9. Evaluate Models

# ==============================
```

```
evaluate_model("Linear Regression", y_test, lr_pred)

evaluate_model("Random Forest", y_test, rf_pred)
```

5. Result

Key observations:

- Linear Regression provided baseline performance.
- XGBoost captured complex patterns more effectively.
- Lag feature (`Subsidy_Lag1`) improved prediction stability.
- Tuned model showed reduced error and better generalization.

6. Conclusion

This project successfully developed a machine learning framework to predict renewable energy transition speed aligned with SDG-7 objectives. Through systematic data engineering, feature creation, and model optimization, the team demonstrated that advanced boosting models significantly outperform traditional linear methods.

The findings highlight the importance of technological cost trends and policy-related variables in accelerating renewable adoption. The developed model can support policymakers and energy planners in monitoring progress toward clean energy goals.

Future Scope:

- Incorporate country-level panel data
- Apply time-series deep learning models
- Build interactive SDG dashboard
- Extend analysis to 2030 forecasting